

Math 340 — Practice Midterm 1

Chapters 1–10 (No Calculator)

Name: _____

Instructions: Show all work neatly in the space provided. Justify your reasoning conceptually where appropriate. Arithmetic should be simple — focus on logic, operations, and structure.

1. Linear Systems and Consistency (Ch 1–2)

(a) Solve the following system using Gaussian elimination:

$$\begin{cases} x + 2y - z = 1, \\ 2x + 3y + z = 4, \\ 3x + 5y + 4z = 9. \end{cases}$$

(b) Write the augmented matrix and determine whether the system is consistent, and whether the solution is unique, infinite, or nonexistent.

(c) Explain the relationship between the number of free variables and the number of solutions in a homogeneous system.

2. Matrix Algebra (Ch 4)

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}.$$

(a) Compute AB and BA . Are they equal?

(b) Show (by computation or reasoning) whether $(A + B)^2 = A^2 + 2AB + B^2$ holds for matrices.

(c) State one property of matrix multiplication that fails which causes (b) to differ from the scalar case.

3. Inverses and Elementary Matrices (Ch 5–6)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

(a) Without using a calculator, find the inverse of A by augmenting with I_3 and performing row operations.

(b) List the sequence of elementary matrices $E_1, E_2, \dots$ such that

$$E_k \cdots E_2 E_1 A = I_3.$$

Explain how these relate to A^{-1} .

4. Determinants and Special Forms (Ch 7–8)

(a) Compute the determinant of the matrix

$$M = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Explain what property of M makes this computation easy.

(b) Find all values of t for which

$$A = \begin{bmatrix} 1 & 2 & t \\ 0 & t & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

is singular.

(c) For a 3×3 matrix A , what does the sign of $\det(A)$ tell you geometrically?

5. Determinants and Row Operations (Ch 9)

Matrix B is obtained from A by the following row operations:

- Swap r_1 and r_2 ,
- Multiply the new r_3 by -2 ,
- Add $3r_1$ to r_3 .

If $\det(B) = 12$, find $\det(A)$.

6. The Adjoint, Inverses, and Homogeneous Systems (Ch 10)

(a) Let

$$C = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute $\det(C^{-1})$ using determinant properties, without finding C^{-1} explicitly.

(b) Determine all values of c for which the homogeneous system

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & c & 3 \\ 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution.

(c) For the matrix in (b), explain conceptually why the determinant condition determines the existence of nontrivial solutions.